Spatial Variation in Childhood Obesity Across London Boroughs

A borough-level comparison of Reception and Year 6 obesity prevalence

Preparation

- Github link:
- Number of words:1325
- Runtime: *0.15hours(colab)*
- Coding environment: *Python 3*
- License: this notebook is made available under the Creative Commons Attribution 4.0 license.
- Additional library:
  - **openpyxl**: used to read the multi-sheet Excel workbook

Table of contents

1. Introduction
2. Research questions
3. Data
4. Methodology
5. Results and discussion
6. Conclusion
7. References

Introduction

[go back to the top]

1.1 Background

Childhood obesity is an important public health problem, which directly damages children's physical and mental health and increases their risk of chronic diseases in the future (World Health Organization, 2025). England conducts routine weight monitoring of Reception and Year 6 children through the National Child Measurement Programme to grasp relevant trends and health inequalities (NHS Digital, 2023). The latest data shows that the obesity rate of Year 6 children is significantly higher than that of Reception children, and health inequalities caused by place and deprivation is still very prominent (Office for Health Improvement and Disparities, 2023).

1.2 Spatial context

From the perspective of spatial data research, childhood obesity is not only a health problem but also reflects the uneven development between regions. By comparing the administrative regions of London, we can see the difference in the obesity rate of children in different regions, and whether these differences vary in different grades. The research data are also very obvious: the obesity rate in London districts is very different. The obesity rate of children in Year 6 has always been higher than that in Reception, and the obesity rate gap between the two grades also varies by district.

1.3 Aim

This study aims to describe the spatial differences in the prevalence of obesity among children in various administrative regions of London, compare the obesity distribution characteristics of Reception and Year 6 children, and analyze whether the degree of poverty at the administrative level is related to the higher prevalence of obesity or the greater difference in school age. The research takes descriptive analysis and spatial analysis as the core, and uses poverty and related models as auxiliary explanation tools.

Research questions

[go back to the top]

2.1 Main research question

What are the spatial differences in the prevalence of childhood obesity in London's administrative regions? What is the difference between these distribution patterns between Reception and Year 6?

2.2 Sub-questions

1.Which districts in London have the highest and lowest rate of child obesity? Is this situation stable in the long run?

2.In which districts is there the biggest obesity rate gap between children in Year 6 and Reception?

3.Will the degree of regional poverty make the obesity rate higher, or the gap between grades larger?

4.After analyzing with the tree model, how much impact do the factors of grade, region, year and poverty have on the obesity rate?

Data

[go back to the top]

3.1 Data source and study area

The main dataset of this study is a childhood obesity workbook at borough level derived from NCMP (NHS Digital, 2023). data link: https://data.london.gov.uk/dataset/prevalence-childhood-obesity-borough

3.2 Data cleaning and restructuring

For the convenience of comparison, only the data from 2007/08 to 2018/19 is retained. Data for 2006/07 because the geographical boundaries are inconsistent, and data for 2019/20 because missing values exist, have been deleted. After that, the multi-row headers were reorganized and only the obesity-related variables was extracted. Finally, the data is sorted into a long table arranged by borough-year-school-stage, and each line corresponds to a London borough, a year and a school stage.

3.3 Analytical variables

The dependent variable of this study is obesity prevalence.The core comparison variable is school_stage, in which boroughs are the spatial analysis units, and year_start is used to describe temporal changes.The study retains confidence intervals for result interpretation, and n_measured is used only for descriptive purposes.The study adds a supplementary variable, imd_score, to reflect the socio-economic background at the borough level.As the data comes from the 2019 deprivation file （data link: https://data.london.gov.uk/dataset/indices-of-deprivation-2l15g/%EF%BC%89, it is regarded as a time-invariant contextual indicator rather than a year-specific variable.

Variable	Type	Description	Notes
`borough`	Categorical	Name of London borough	Spatial unit of analysis
`year_label`	Categorical	Academic year label (e.g. `2018-19`)	Used for filtering and display
`year_start`	Numeric	First year of the academic year (e.g. `2018`)	Used as an ordered time variable
`school_stage`	Categorical	`Reception` or `Year 6`	Main non-spatial comparison variable
`obesity_prevalence`	Numeric	Obesity prevalence (%)	Main dependent variable
`obesity_ci_lower`	Numeric	Lower confidence interval	Retained for interpretation
`obesity_ci_upper`	Numeric	Upper confidence interval	Retained for interpretation

Variable	Type	Description	Notes
`n_measured`	Numeric	Number of children measured	Descriptive only; not the main analytical focus

Methodology

[go back to the top]

4.1 Analytical workflow

This analysis conducts spatial analysis of London's boroughs. First, it cleans and reshapes the data, then uses descriptive statistics and visualization to compare obesity data across different boroughs, school stages and years. Through spatial visualization, the obesity prevalence distribution in the 2018/19 academic year is presented, differences in obesity rates between school stages are analyzed, and poverty analysis and tree-based models are used to aid interpretation.

4.2 Supplementary deprivation and modelling methods

The study uses the IMD deprivation index across London boroughs to analyze the relationship between poverty, obesity prevalence, and school-stage obesity gaps, with tests using scatter plots and correlation coefficients. At the same time, regression tree and random forest models are constructed, with obesity prevalence as the dependent variable and borough, school_stage, year, and IMD score as independent variables. The model is evaluated using R^2 and RMSE, and results are interpreted using permutation importance.

In [3]:
```
# ========================
# Imports
# ========================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Optional / used later
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor, plot_tree
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.inspection import permutation_importance

# Display options
pd.set_option("display.max_columns", 100)
pd.set_option("display.width", 120)
```

In [2]:
```
# ========================
# Data import and cleaning
# ========================

file_path = "childhood-obesity-borough.xlsx"
```

```
LONDON_BOROUGHS = [
    'Barking and Dagenham', 'Barnet', 'Bexley', 'Brent', 'Bromley', 'Camden',
    'Croydon', 'Ealing', 'Enfield', 'Greenwich', 'Hammersmith and Fulham',
    'Haringey', 'Harrow', 'Havering', 'Hillingdon', 'Hounslow', 'Islington',
    'Kensington and Chelsea', 'Kingston upon Thames', 'Lambeth', 'Lewisham',
    'Merton', 'Newham', 'Redbridge', 'Richmond upon Thames', 'Southwark',
    'Sutton', 'Tower Hamlets', 'Waltham Forest', 'Wandsworth', 'Westminster'
]

YEARS = [f"{y}-{str(y+1)[-2:]}" for y in range(2007, 2019)]

def extract_obesity_sheet(file_path, sheet_name):
    raw = pd.read_excel(file_path, sheet_name=sheet_name, header=None)

    header_rows = raw.iloc[:3].copy().ffill(axis=1)

    col_names = []
    for j in range(raw.shape[1]):
        parts = [
            str(header_rows.iloc[i, j]).strip()
            for i in range(3)
            if pd.notna(header_rows.iloc[i, j])
        ]
        col_names.append(" | ".join(parts))

    df = raw.iloc[4:].copy()
    df.columns = col_names

    df = df.rename(columns={
        col_names[0]: "code",
        col_names[1]: "borough"
    })

    df = df[df["borough"].isin(LONDON_BOROUGHS)].copy()

    obesity_prefix = None
    for c in df.columns:
        c_low = c.lower()
        if ("obese" in c_low) and ("prevalence" in c_low) and ("severely obese"
            obesity_prefix = c.split(" | ")[0]
            break

    if obesity_prefix is None:
        raise ValueError(f"Could not find obesity columns in sheet {sheet_name}"

    def pick_col(stage, keywords):
        for c in df.columns:
            c_low = c.lower()
            if c.startswith(obesity_prefix) and stage.lower() in c_low and all(k
                return c
        raise ValueError(f"Missing column for {stage} / {keywords} in sheet {she

    rec_prev = pick_col("Reception", ["prevalence"])
    y6_prev = pick_col("Year 6", ["prevalence"])

    has_lower_upper = any(
        c.startswith(obesity_prefix) and "lower confidence interval" in c.lower(
        for c in df.columns
    )
```

```
    measured_cols = [c for c in df.columns if "number of children measured" in c
    rec_n = [c for c in measured_cols if "reception" in c.lower()][0]
    y6_n = [c for c in measured_cols if "year 6" in c.lower()][0]

    out = pd.DataFrame({
        "borough": df["borough"],
        "year_label": sheet_name,
        "year_start": int(sheet_name[:4]),
        "reception_obesity_prevalence": pd.to_numeric(df[rec_prev], errors="coer
        "year6_obesity_prevalence": pd.to_numeric(df[y6_prev], errors="coerce"),
        "reception_n_measured": pd.to_numeric(df[rec_n], errors="coerce"),
        "year6_n_measured": pd.to_numeric(df[y6_n], errors="coerce"),
    })

    if has_lower_upper:
        rec_low = pick_col("Reception", ["lower confidence interval"])
        rec_up  = pick_col("Reception", ["upper confidence interval"])
        y6_low  = pick_col("Year 6", ["lower confidence interval"])
        y6_up   = pick_col("Year 6", ["upper confidence interval"])

        out["reception_ci_lower"] = pd.to_numeric(df[rec_low], errors="coerce")
        out["reception_ci_upper"] = pd.to_numeric(df[rec_up], errors="coerce")
        out["year6_ci_lower"]     = pd.to_numeric(df[y6_low], errors="coerce")
        out["year6_ci_upper"]     = pd.to_numeric(df[y6_up], errors="coerce")
    else:
        rec_margin = pick_col("Reception", ["confidence interval"])
        y6_margin  = pick_col("Year 6", ["confidence interval"])

        rec_margin_num = pd.to_numeric(df[rec_margin], errors="coerce")
        y6_margin_num  = pd.to_numeric(df[y6_margin], errors="coerce")

        out["reception_ci_lower"] = out["reception_obesity_prevalence"] - rec_ma
        out["reception_ci_upper"] = out["reception_obesity_prevalence"] + rec_ma
        out["year6_ci_lower"]     = out["year6_obesity_prevalence"] - y6_margin_
        out["year6_ci_upper"]     = out["year6_obesity_prevalence"] + y6_margin_

    return out

wide_parts = [extract_obesity_sheet(file_path, year) for year in YEARS]
obesity_wide = pd.concat(wide_parts, ignore_index=True)

reception_df = (
    obesity_wide[
        ["borough", "year_label", "year_start",
         "reception_obesity_prevalence", "reception_ci_lower", "reception_ci_upp
    ]
    .rename(columns={
        "reception_obesity_prevalence": "obesity_prevalence",
        "reception_ci_lower": "obesity_ci_lower",
        "reception_ci_upper": "obesity_ci_upper",
        "reception_n_measured": "n_measured"
    })
    .assign(school_stage="Reception")
)

year6_df = (
    obesity_wide[
        ["borough", "year_label", "year_start",
         "year6_obesity_prevalence", "year6_ci_lower", "year6_ci_upper", "year6_
```

```
    ]
    .rename(columns={
        "year6_obesity_prevalence": "obesity_prevalence",
        "year6_ci_lower": "obesity_ci_lower",
        "year6_ci_upper": "obesity_ci_upper",
        "year6_n_measured": "n_measured"
    })
    .assign(school_stage="Year 6")
)

obesity_model_long = pd.concat([reception_df, year6_df], ignore_index=True)

obesity_model_long = obesity_model_long[
    ["borough", "year_label", "year_start", "school_stage",
     "obesity_prevalence", "obesity_ci_lower", "obesity_ci_upper", "n_measured"]
].copy()

obesity_model_long["borough"] = obesity_model_long["borough"].astype(str).str.st
obesity_model_long["year_label"] = obesity_model_long["year_label"].astype("cate
obesity_model_long["school_stage"] = pd.Categorical(
    obesity_model_long["school_stage"],
    categories=["Reception", "Year 6"],
    ordered=True
)

obesity_map_2018 = obesity_model_long[obesity_model_long["year_label"] == "2018-

print("obesity_wide shape:", obesity_wide.shape)
print("obesity_model_long shape:", obesity_model_long.shape)
print("obesity_map_2018 shape:", obesity_map_2018.shape)

display(obesity_model_long.head())

obesity_model_long.to_csv("obesity_model_long.csv", index=False)
obesity_map_2018.to_csv("obesity_map_2018.csv", index=False)
```

```
obesity_wide shape: (372, 11)
obesity_model_long shape: (744, 8)
obesity_map_2018 shape: (62, 8)
```

	borough	year_label	year_start	school_stage	obesity_prevalence	obesity_ci_low
0	Barking and Dagenham	2007-08	2007	Reception	13.492823	12.0277
1	Barnet	2007-08	2007	Reception	8.846283	7.8253
2	Bexley	2007-08	2007	Reception	9.604520	8.3511
3	Brent	2007-08	2007	Reception	10.639021	9.5246
4	Bromley	2007-08	2007	Reception	7.306684	6.3832

In [4]:

```
# ========================
# Build borough-level deprivation data from ward-level file
# ========================

dep_file = "London wards id2019 summary measures.xlsx"
```

```
dep_sheet = "wards_id_summary"

dep_raw = pd.read_excel(dep_file, sheet_name=dep_sheet)

dep_raw = dep_raw[["Borough", "IMD average score"]].copy()
dep_raw["Borough"] = dep_raw["Borough"].astype(str).str.strip()

borough_dep = (
    dep_raw.groupby("Borough", as_index=False)["IMD average score"]
    .mean()
    .rename(columns={
        "Borough": "borough",
        "IMD average score": "imd_score"
    })
)

borough_dep["borough"] = borough_dep["borough"].astype(str).str.strip()

# Merge into main table
obesity_model_long_enhanced = obesity_model_long.merge(
    borough_dep,
    on="borough",
    how="left"
)

missing_boroughs = (
    obesity_model_long_enhanced.loc[
        obesity_model_long_enhanced["imd_score"].isna(), ["borough"]
    ]
    .drop_duplicates()
    .sort_values("borough")
)

print("Missing deprivation values:", obesity_model_long_enhanced["imd_score"].is
if len(missing_boroughs) > 0:
    display(missing_boroughs)

display(borough_dep.head())
display(obesity_model_long_enhanced.head())

obesity_model_long_enhanced.to_csv("obesity_model_long_enhanced.csv", index=Fals
borough_dep.to_csv("borough_deprivation_from_ward_imd2019.csv", index=False)
```

```
Missing deprivation values: 0
```

	borough	imd_score
0	Barking and Dagenham	32.768253
1	Barnet	15.821108
2	Bexley	16.473071
3	Brent	25.095808
4	Bromley	14.041901

	borough	year_label	year_start	school_stage	obesity_prevalence	obesity_ci_low
0	Barking and Dagenham	2007-08	2007	Reception	13.492823	12.0277
1	Barnet	2007-08	2007	Reception	8.846283	7.8253
2	Bexley	2007-08	2007	Reception	9.604520	8.3511
3	Brent	2007-08	2007	Reception	10.639021	9.5246
4	Bromley	2007-08	2007	Reception	7.306684	6.3832

In [5]:

```
# ========================
# Descriptive summary
# ========================

summary_by_stage = (
    obesity_model_long.groupby("school_stage", observed=False)["obesity_prevalen
    .agg(["count", "mean", "median", "min", "max", "std"])
    .round(2)
)

summary_by_year_stage = (
    obesity_model_long.groupby(["year_label", "school_stage"], observed=False)["
    .mean()
    .reset_index()
)

summary_by_borough_stage = (
    obesity_model_long.groupby(["borough", "school_stage"], observed=False)["obe
    .mean()
    .reset_index()
)

display(summary_by_stage)
display(summary_by_year_stage.head())
display(summary_by_borough_stage.head())
```

school_stage	count	mean	median	min	max	std
Reception	372	10.37	10.41	4.59	14.74	2.1
Year 6	372	22.06	22.66	10.67	30.03	3.5

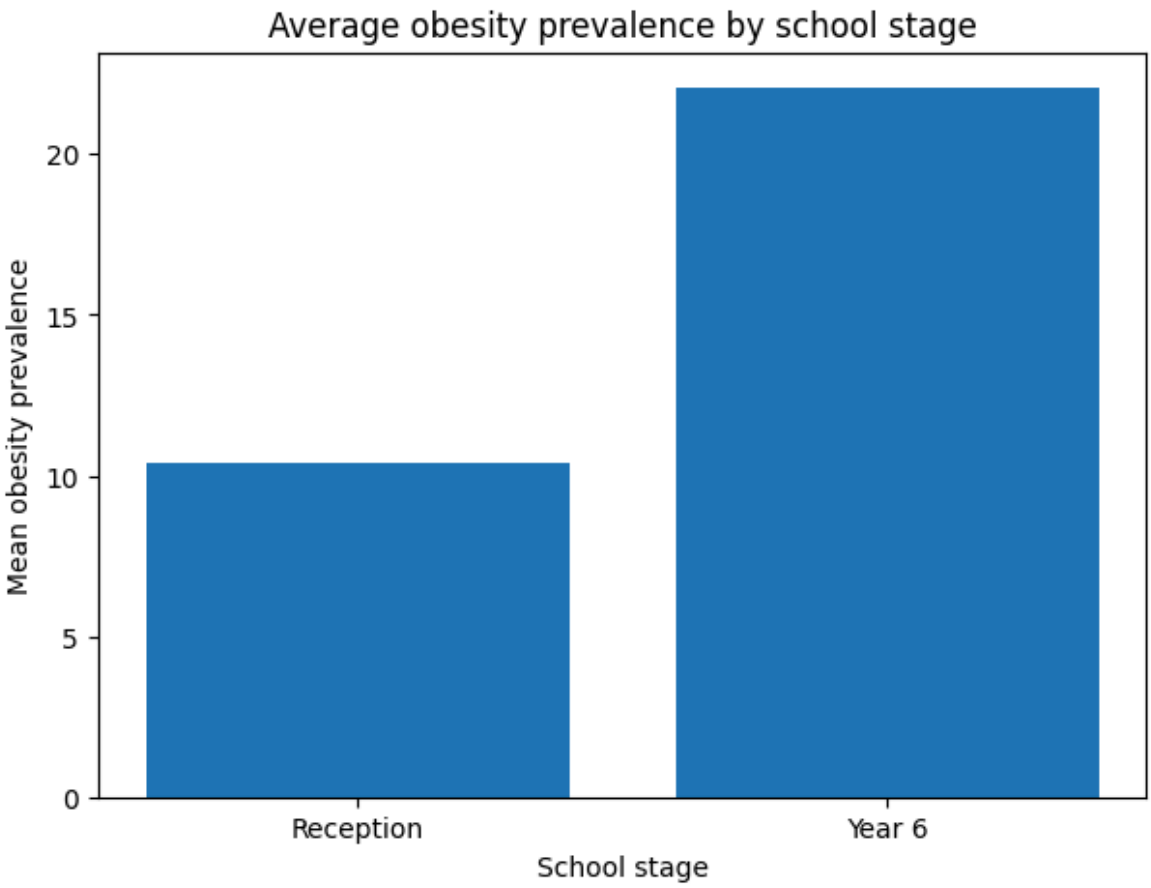
	year_label	school_stage	obesity_prevalence
0	2007-08	Reception	10.683106
1	2007-08	Year 6	21.378639
2	2008-09	Reception	11.008643
3	2008-09	Year 6	21.023052
4	2009-10	Reception	11.252451

	borough	school_stage	obesity_prevalence
0	Barking and Dagenham	Reception	13.465471
1	Barking and Dagenham	Year 6	26.364744
2	Barnet	Reception	9.259206
3	Barnet	Year 6	19.036468
4	Bexley	Reception	10.967543

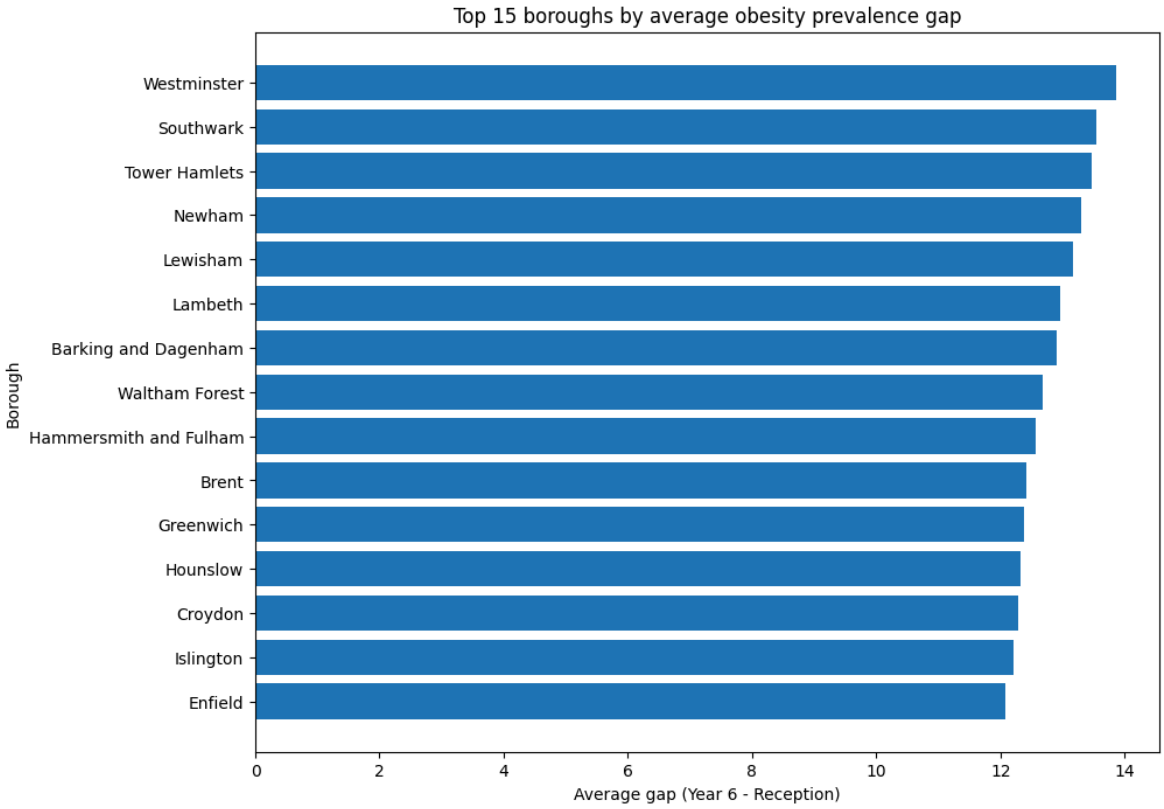
In [6]:

```
# ========================
# Compare Reception and Year 6
# ========================

stage_means = (
    obesity_model_long.groupby("school_stage", observed=False)["obesity_prevalen
    .mean()
    .reset_index()
)

plt.figure(figsize=(7, 5))
plt.bar(stage_means["school_stage"], stage_means["obesity_prevalence"])
plt.xlabel("School stage")
plt.ylabel("Mean obesity prevalence")
plt.title("Average obesity prevalence by school stage")
plt.show()

borough_stage_pivot = (
    summary_by_borough_stage.pivot(index="borough", columns="school_stage", valu
    .dropna()
)
borough_stage_pivot["gap_y6_minus_reception"] = borough_stage_pivot["Year 6"] -

gap_rank_all_years = (
    borough_stage_pivot[["gap_y6_minus_reception"]]
    .sort_values("gap_y6_minus_reception", ascending=False)
    .reset_index()
)

display(gap_rank_all_years.head(10))

plt.figure(figsize=(10, 8))
top_gap = gap_rank_all_years.head(15).sort_values("gap_y6_minus_reception", asce
plt.barh(top_gap["borough"], top_gap["gap_y6_minus_reception"])
plt.xlabel("Average gap (Year 6 - Reception)")
plt.ylabel("Borough")
```

```
plt.title("Top 15 boroughs by average obesity prevalence gap")
plt.show()
```

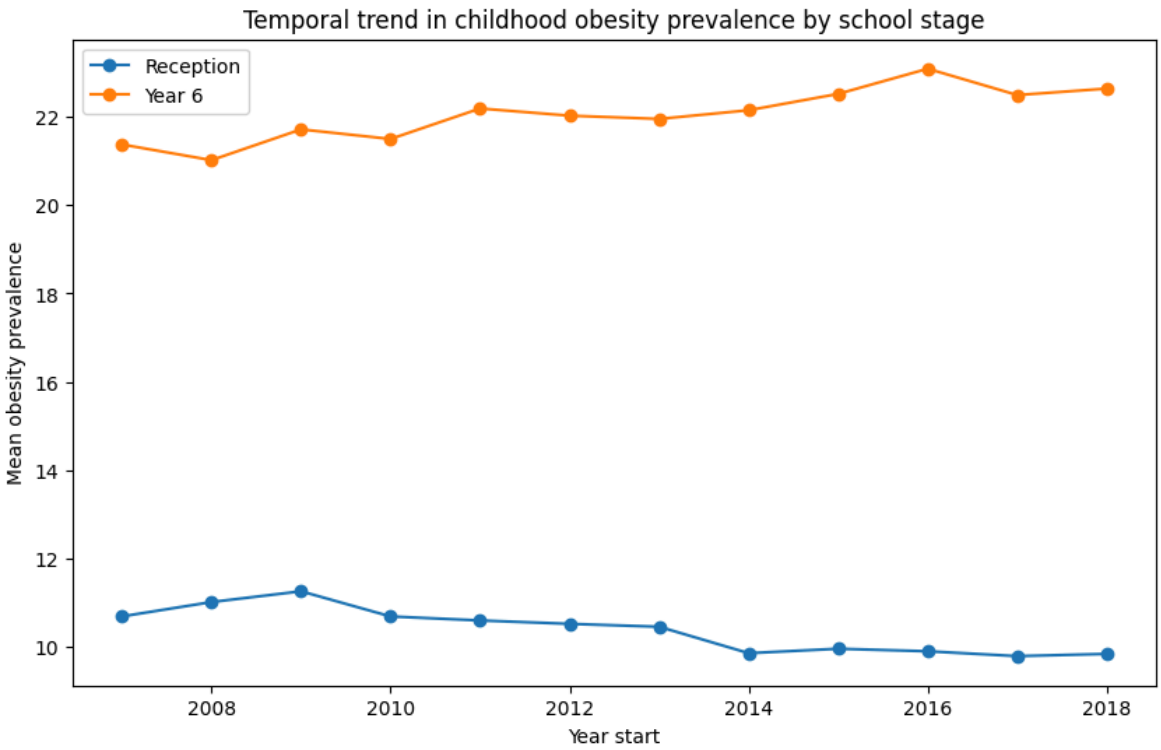


school_stage	borough	gap_y6_minus_reception
0	Westminster	13.865640
1	Southwark	13.550022
2	Tower Hamlets	13.461276
3	Newham	13.299456
4	Lewisham	13.167725
5	Lambeth	12.965401
6	Barking and Dagenham	12.899273
7	Waltham Forest	12.687521
8	Hammersmith and Fulham	12.563859
9	Brent	12.424492



In [7]:

```
# ========================
# Temporal trends by school stage
# ========================

trend_df = (
    obesity_model_long.groupby(["year_start", "school_stage"], observed=False)["
    .mean()
    .reset_index()
)

plt.figure(figsize=(10, 6))
for stage in ["Reception", "Year 6"]:
    stage_data = trend_df[trend_df["school_stage"] == stage]
    plt.plot(stage_data["year_start"], stage_data["obesity_prevalence"], marker=

plt.xlabel("Year start")
plt.ylabel("Mean obesity prevalence")
plt.title("Temporal trend in childhood obesity prevalence by school stage")
plt.legend()
plt.show()
```



In [8]:

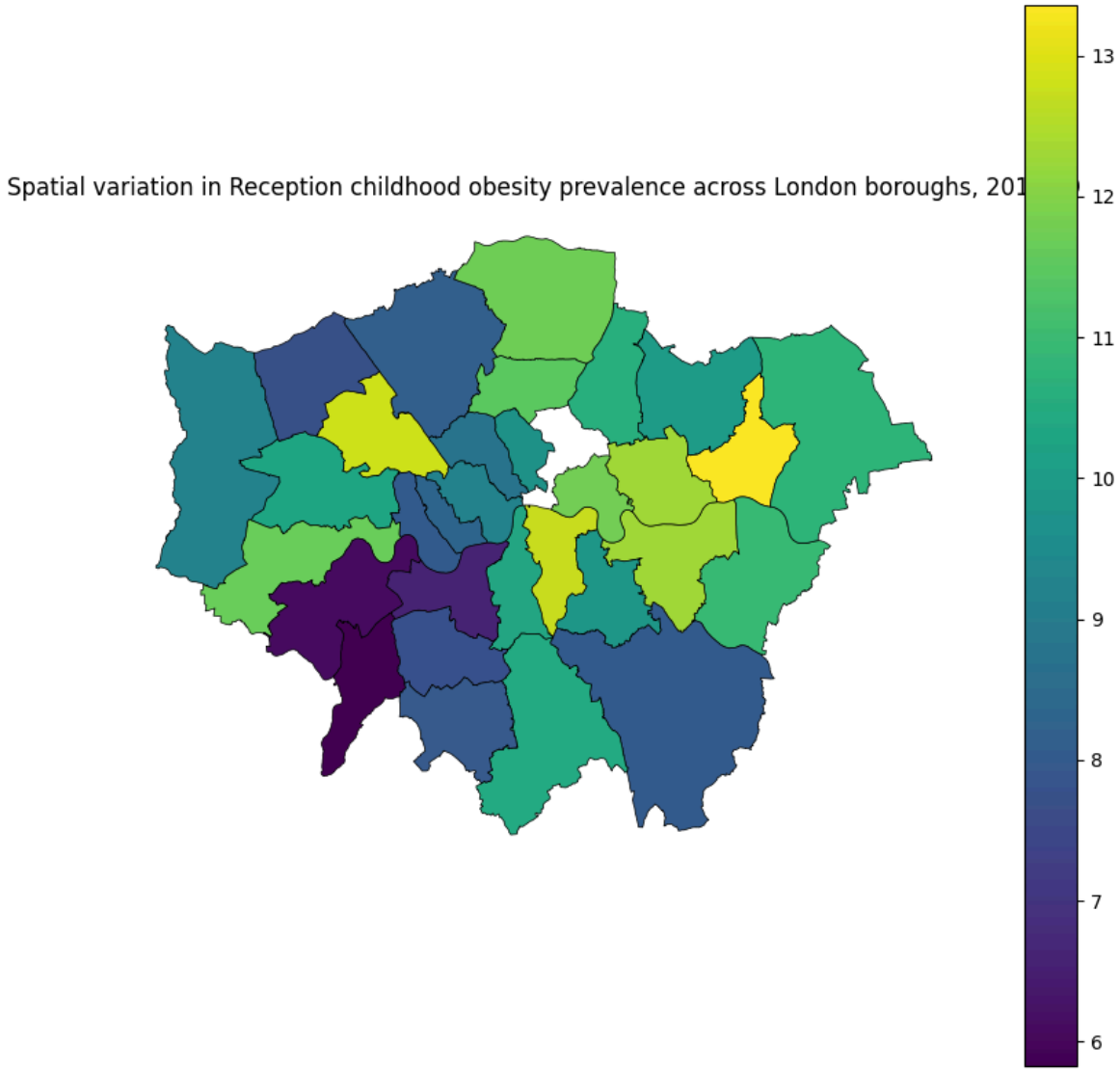
```
# ========================
# London borough choropleth maps
# ========================

import geopandas as gpd

# Public GeoJSON source
geojson_url = "https://raw.githubusercontent.com/radoi90/housequest-data/master/
borough_gdf = gpd.read_file(geojson_url)

# Find borough name column
name_col = None
for candidate in ["name", "Name", "NAME", "borough", "Borough"]:
    if candidate in borough_gdf.columns:
        name_col = candidate
        break

if name_col is None:
    raise ValueError("Could not find borough name column in the GeoJSON file.")

borough_gdf = borough_gdf.rename(columns={name_col: "borough"})
borough_gdf["borough"] = borough_gdf["borough"].astype(str).str.strip()

# Keep only boroughs in the obesity dataset
valid_boroughs = set(obesity_map_2018["borough"].unique())
borough_gdf = borough_gdf[borough_gdf["borough"].isin(valid_boroughs)].copy()

map_df = obesity_map_2018.copy()
map_df["obesity_prevalence"] = pd.to_numeric(map_df["obesity_prevalence"], error

reception_map = map_df[map_df["school_stage"] == "Reception"][["borough", "obesi
year6_map = map_df[map_df["school_stage"] == "Year 6"][["borough", "obesity_prev

reception_geo = borough_gdf.merge(reception_map, on="borough", how="left")
year6_geo = borough_gdf.merge(year6_map, on="borough", how="left")
```

```
gap_df_2018 = (
    map_df.pivot(index="borough", columns="school_stage", values="obesity_preval
    .reset_index()
    .dropna()
)
gap_df_2018["gap_y6_minus_reception"] = gap_df_2018["Year 6"] - gap_df_2018["Rec

gap_geo = borough_gdf.merge(
    gap_df_2018[["borough", "gap_y6_minus_reception"]],
    on="borough",
    how="left"
)

print("Reception map rows:", reception_geo.shape[0])
print("Year 6 map rows:", year6_geo.shape[0])
print("Gap map rows:", gap_geo.shape[0])
```

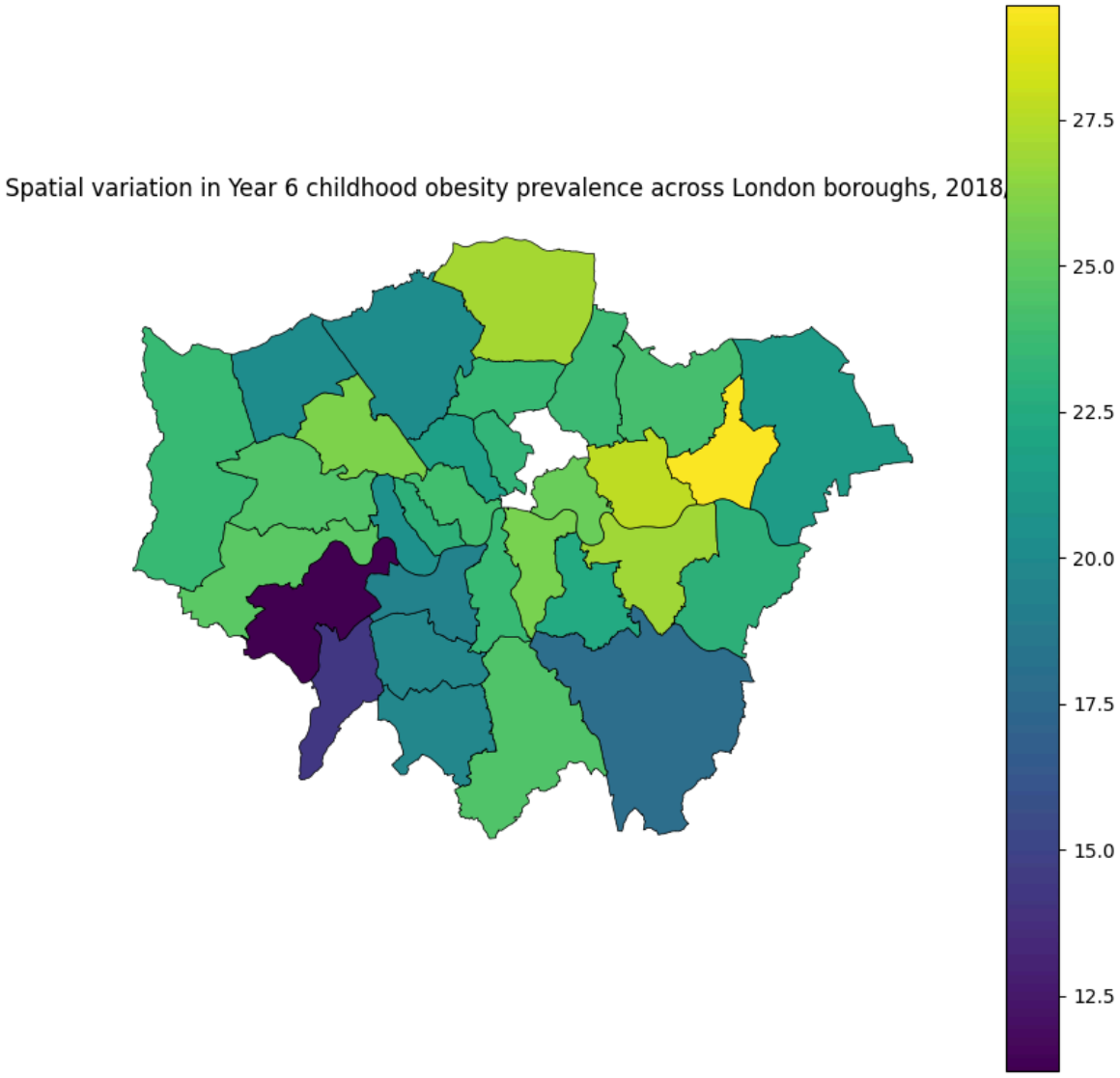
```
Reception map rows: 31
Year 6 map rows: 31
Gap map rows: 31
```

In [9]:

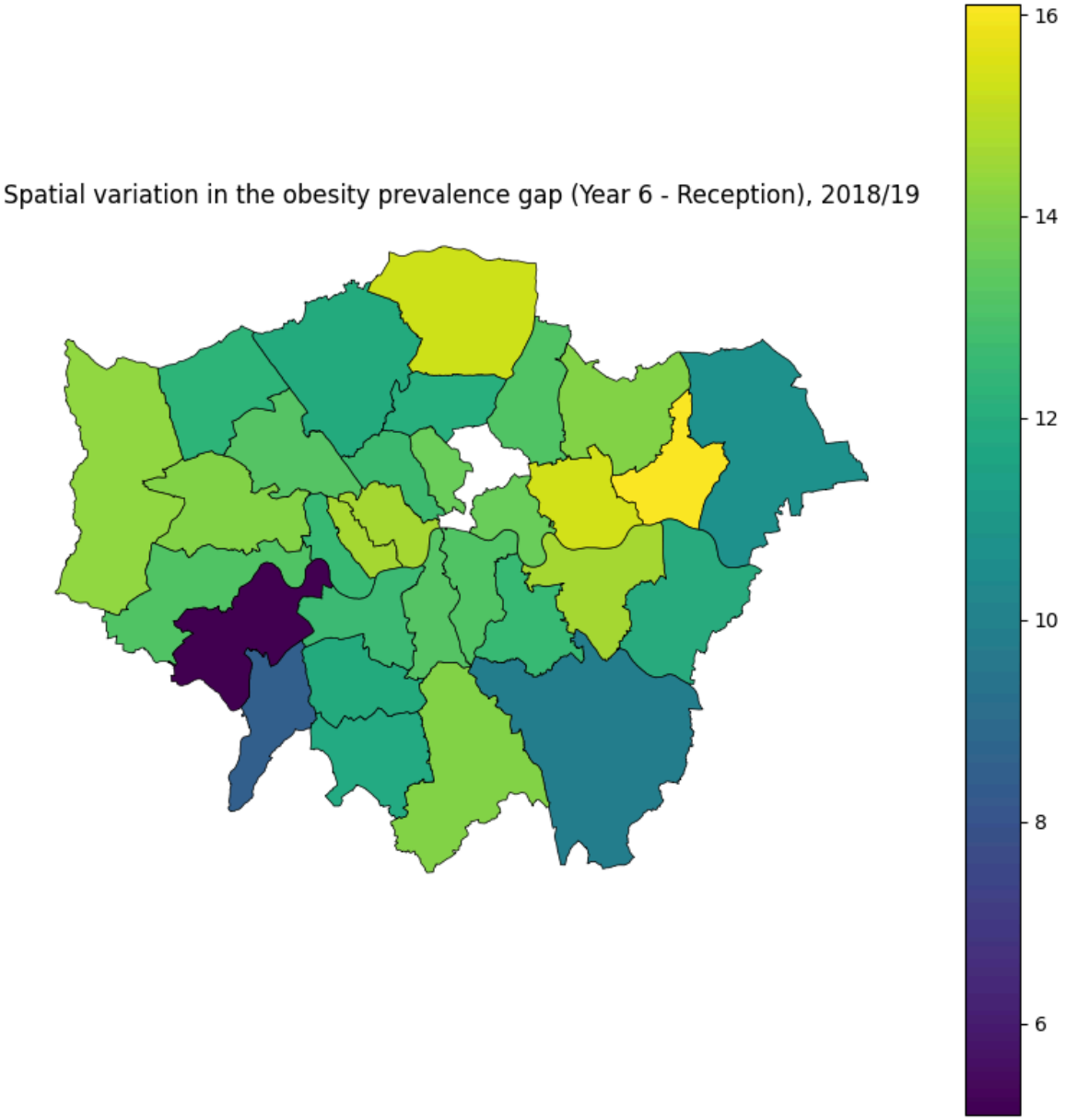
```
# Plot: Reception obesity prevalence map
fig, ax = plt.subplots(figsize=(10, 10))
reception_geo.plot(
    column="obesity_prevalence",
    ax=ax,
    legend=True,
    edgecolor="black",
    linewidth=0.5,
    missing_kwds={"color": "lightgrey", "label": "Missing"}
)
ax.set_title("Spatial variation in Reception childhood obesity prevalence across
ax.axis("off")
plt.show()
```



In [10]:
```
# Plot: Year 6 obesity prevalence map
fig, ax = plt.subplots(figsize=(10, 10))
year6_geo.plot(
    column="obesity_prevalence",
    ax=ax,
    legend=True,
    edgecolor="black",
    linewidth=0.5,
    missing_kwds={"color": "lightgrey", "label": "Missing"}
)
ax.set_title("Spatial variation in Year 6 childhood obesity prevalence across Lo
ax.axis("off")
plt.show()
```



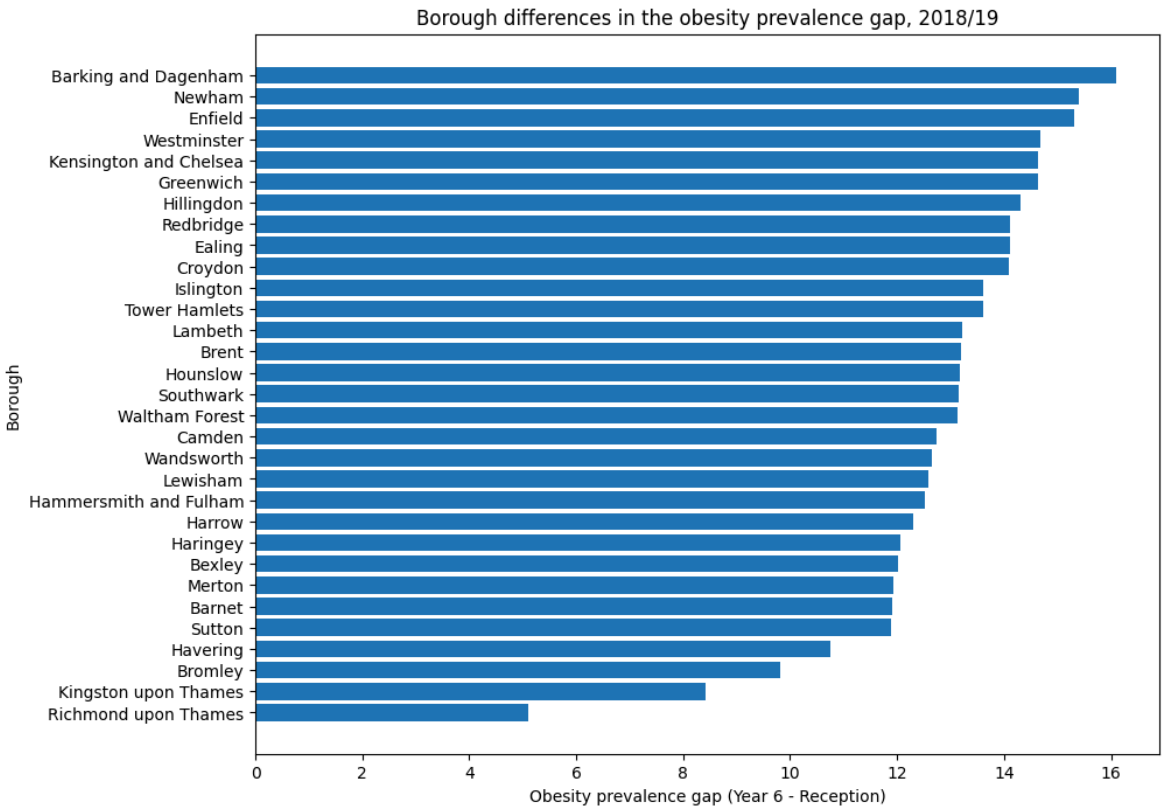
In [11]:
```
# Plot: Year 6 - Reception gap map
fig, ax = plt.subplots(figsize=(10, 10))
gap_geo.plot(
    column="gap_y6_minus_reception",
    ax=ax,
    legend=True,
    edgecolor="black",
    linewidth=0.5,
    missing_kwds={"color": "lightgrey", "label": "Missing"}
)
ax.set_title("Spatial variation in the obesity prevalence gap (Year 6 - Receptio
ax.axis("off")
plt.show()
```



In [12]:

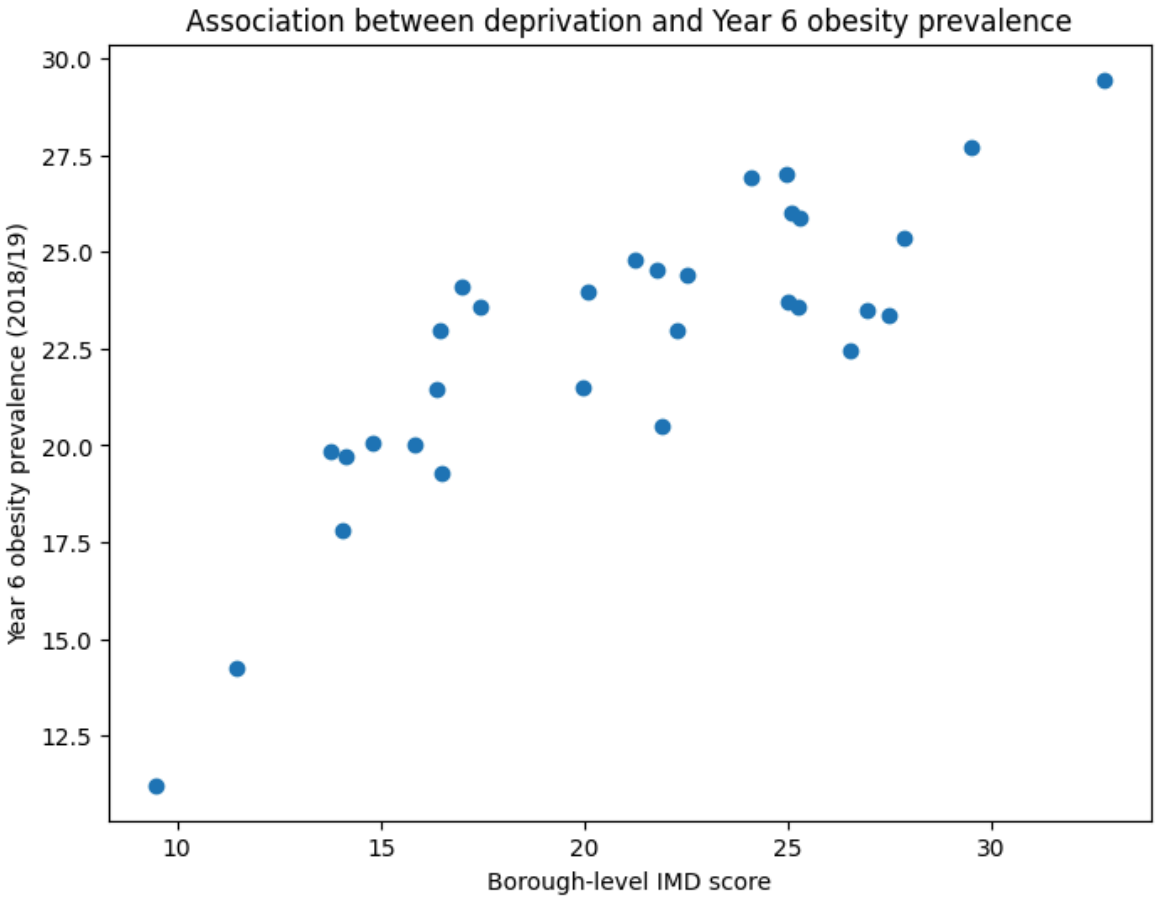
```
# ========================
# Gap ranking for 2018/19
# ========================

gap_rank_2018 = gap_df_2018[["borough", "gap_y6_minus_reception"]].sort_values(
    "gap_y6_minus_reception", ascending=False
)

display(gap_rank_2018.head(10))

plt.figure(figsize=(10, 8))
gap_rank_plot = gap_rank_2018.sort_values("gap_y6_minus_reception", ascending=Tr
plt.barh(gap_rank_plot["borough"], gap_rank_plot["gap_y6_minus_reception"])
plt.xlabel("Obesity prevalence gap (Year 6 - Reception)")
plt.ylabel("Borough")
plt.title("Borough differences in the obesity prevalence gap, 2018/19")
plt.show()
```

school_stage	borough	gap_y6_minus_reception
0	Barking and Dagenham	16.100413
22	Newham	15.410073
8	Enfield	15.314136
30	Westminster	14.679282
17	Kensington and Chelsea	14.630757
9	Greenwich	14.630468
14	Hillingdon	14.314172
23	Redbridge	14.109015
7	Ealing	14.103615
6	Croydon	14.085560



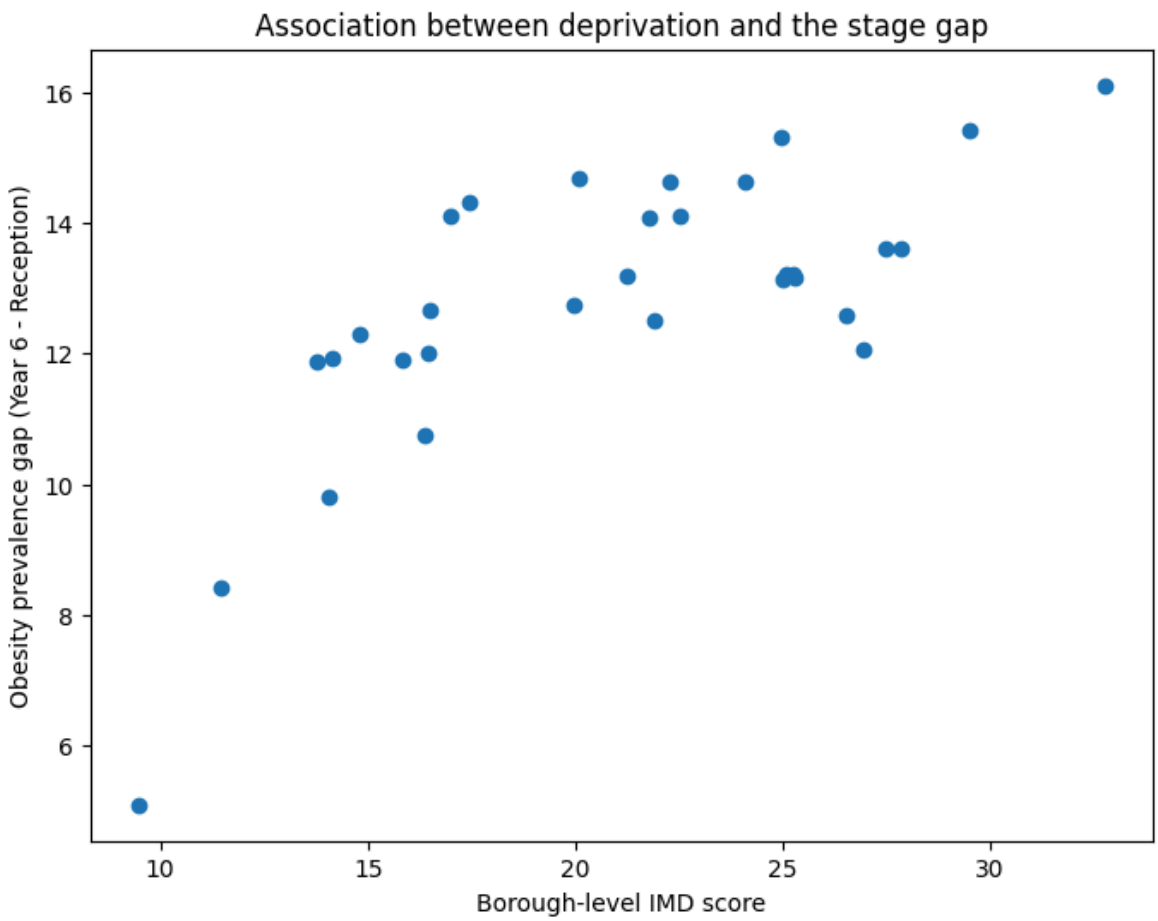
In [13]:
```
# 2018 Year 6 obesity + deprivation
year6_dep = year6_map.merge(borough_dep, on="borough", how="left")

plt.figure(figsize=(8, 6))
plt.scatter(year6_dep["imd_score"], year6_dep["obesity_prevalence"])
plt.xlabel("Borough-level IMD score")
plt.ylabel("Year 6 obesity prevalence (2018/19)")
plt.title("Association between deprivation and Year 6 obesity prevalence")
plt.show()

print("Correlation: IMD score vs Year 6 obesity prevalence")
display(year6_dep[["imd_score", "obesity_prevalence"]].corr())

# 2018 gap + deprivation
gap_dep = gap_df_2018.merge(borough_dep, on="borough", how="left")
```

```
plt.figure(figsize=(8, 6))
plt.scatter(gap_dep["imd_score"], gap_dep["gap_y6_minus_reception"])
plt.xlabel("Borough-level IMD score")
plt.ylabel("Obesity prevalence gap (Year 6 - Reception)")
plt.title("Association between deprivation and the stage gap")
plt.show()

print("Correlation: IMD score vs stage gap")
display(gap_dep[["imd_score", "gap_y6_minus_reception"]].corr())
```




```
Correlation: IMD score vs Year 6 obesity prevalence
```

	imd_score	obesity_prevalence
imd_score	1.000000	0.832241
obesity_prevalence	0.832241	1.000000



Correlation: IMD score vs stage gap

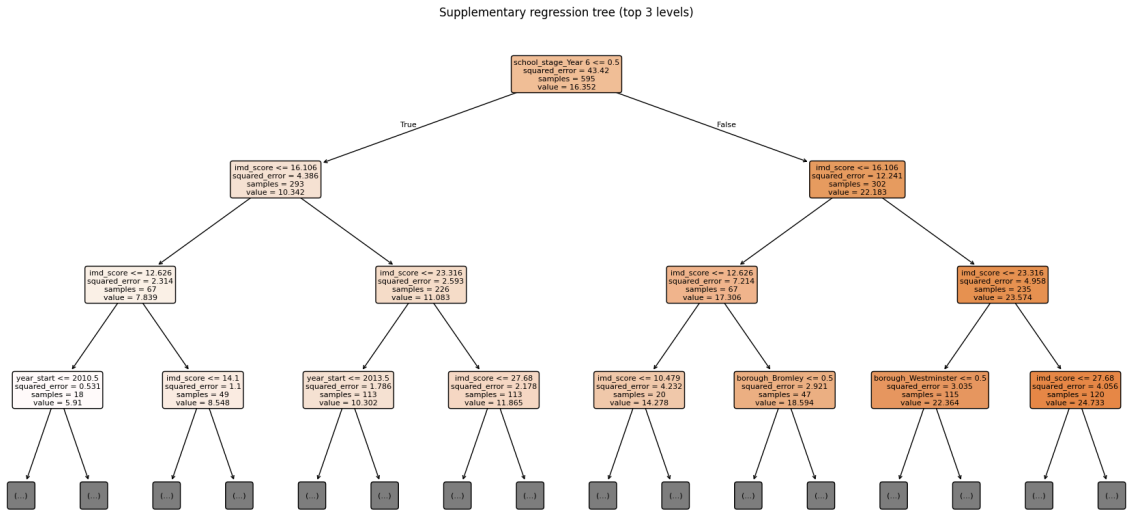
	imd_score	gap_y6_minus_reception
imd_score	1.000000	0.715558
gap_y6_minus_reception	0.715558	1.000000

In [14]:

```
# ========================
# Supplementary regression tree model
# ========================

model_df = obesity_model_long_enhanced[
    ["borough", "school_stage", "year_start", "imd_score", "obesity_prevalence"]
].dropna().copy()

tree_X = pd.get_dummies(
    model_df[["borough", "school_stage", "year_start", "imd_score"]],
    columns=["borough", "school_stage"],
    drop_first=True
)
tree_y = model_df["obesity_prevalence"].astype(float)

tree_X_train, tree_X_test, tree_y_train, tree_y_test = train_test_split(
    tree_X, tree_y, test_size=0.2, random_state=42
)

tree_model = DecisionTreeRegressor(
    max_depth=4,
    min_samples_leaf=4,
    random_state=42
)
```

```python
tree_model.fit(tree_X_train, tree_y_train)

tree_pred = tree_model.predict(tree_X_test)

tree_r2 = r2_score(tree_y_test, tree_pred)
tree_rmse = np.sqrt(mean_squared_error(tree_y_test, tree_pred))

print(f"Supplementary Regression Tree Test R²: {tree_r2:.3f}")
print(f"Supplementary Regression Tree Test RMSE: {tree_rmse:.3f}")

plt.figure(figsize=(22, 10))
plot_tree(
    tree_model,
    feature_names=tree_X.columns,
    filled=True,
    rounded=True,
    fontsize=8,
    max_depth=3
)
plt.title("Supplementary regression tree (top 3 levels)")
plt.show()
```

```
Supplementary Regression Tree Test R²: 0.945
Supplementary Regression Tree Test RMSE: 1.449
```



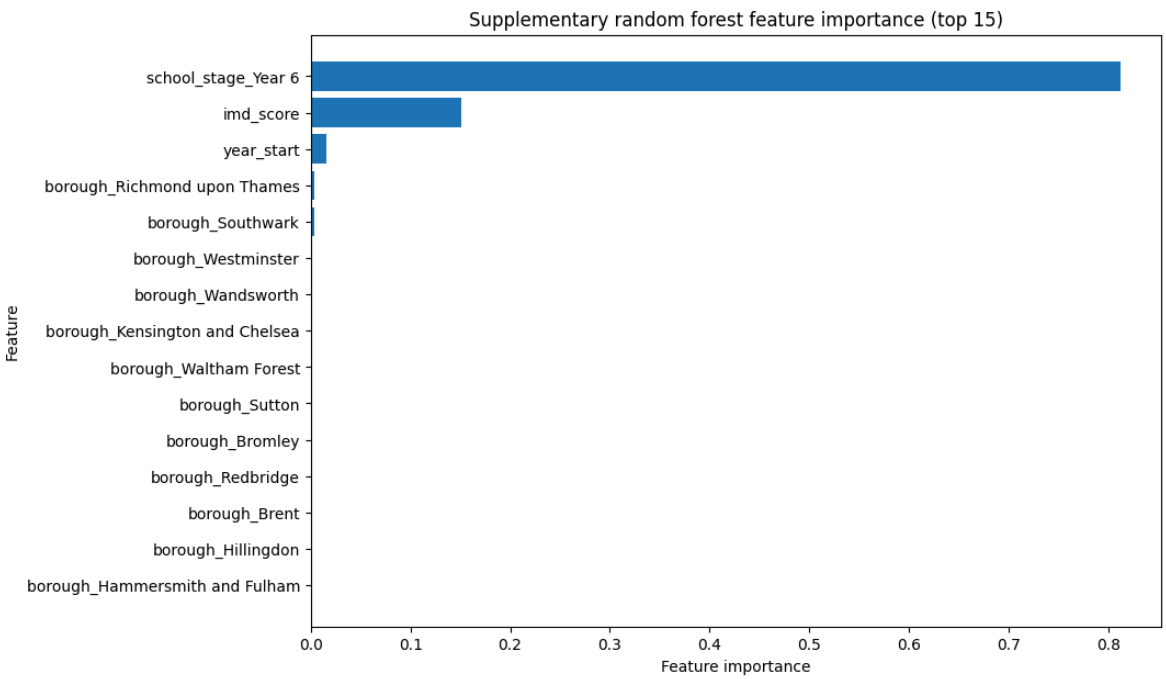
In [15]:

```python
# ========================
# Supplementary random forest model
# ========================

model_df = obesity_model_long_enhanced[
    ["borough", "school_stage", "year_start", "imd_score", "obesity_prevalence"]
].dropna().copy()

rf_X = pd.get_dummies(
    model_df[["borough", "school_stage", "year_start", "imd_score"]],
    columns=["borough", "school_stage"],
    drop_first=True
)
rf_y = model_df["obesity_prevalence"].astype(float)

rf_X_train, rf_X_test, rf_y_train, rf_y_test = train_test_split(
    rf_X, rf_y, test_size=0.2, random_state=42
)
```

```python
rf_model = RandomForestRegressor(
    n_estimators=300,
    max_depth=None,
    min_samples_leaf=2,
    random_state=42
)
rf_model.fit(rf_X_train, rf_y_train)

rf_pred = rf_model.predict(rf_X_test)

rf_r2 = r2_score(rf_y_test, rf_pred)
rf_rmse = np.sqrt(mean_squared_error(rf_y_test, rf_pred))

print(f"Supplementary RF Test R²: {rf_r2:.3f}")
print(f"Supplementary RF Test RMSE: {rf_rmse:.3f}")

importance_df = pd.DataFrame({
    "feature": rf_X.columns,
    "importance": rf_model.feature_importances_
}).sort_values("importance", ascending=False)

display(importance_df.head(15).reset_index(drop=True))

top15 = importance_df.head(15).sort_values("importance", ascending=True)

plt.figure(figsize=(10, 7))
plt.barh(top15["feature"], top15["importance"])
plt.xlabel("Feature importance")
plt.ylabel("Feature")
plt.title("Supplementary random forest feature importance (top 15)")
plt.show()
```

```
Supplementary RF Test R²: 0.972
Supplementary RF Test RMSE: 1.032
```

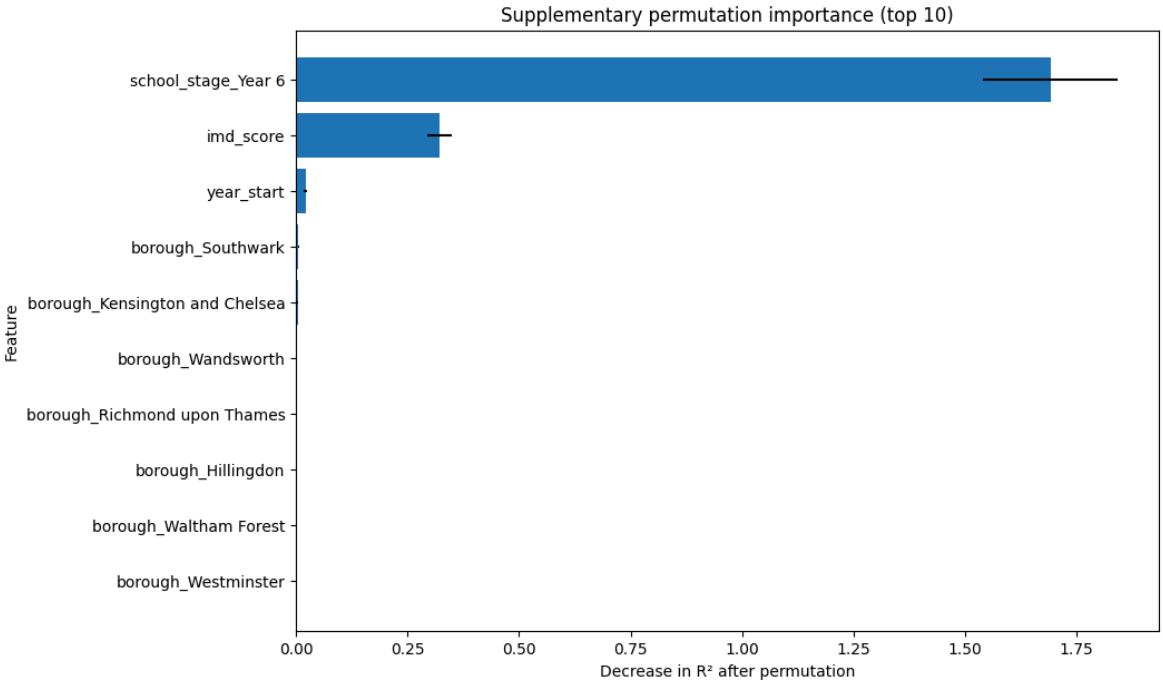
	feature	importance
0	school_stage_Year 6	0.812315
1	imd_score	0.150914
2	year_start	0.015299
3	borough_Richmond upon Thames	0.003489
4	borough_Southwark	0.003168
5	borough_Westminster	0.001786
6	borough_Wandsworth	0.001362
7	borough_Kensington and Chelsea	0.001127
8	borough_Waltham Forest	0.000999
9	borough_Sutton	0.000832
10	borough_Bromley	0.000771
11	borough_Redbridge	0.000739
12	borough_Brent	0.000708
13	borough_Hillingdon	0.000650
14	borough_Hammersmith and Fulham	0.000637



In [16]:
```
# ========================
# Supplementary permutation importance
# ========================

perm_result = permutation_importance(
    estimator=rf_model,
    X=rf_X_test,
    y=rf_y_test,
    n_repeats=20,
    random_state=42,
```

```
    scoring="r2",
    n_jobs=-1
)

perm_df = pd.DataFrame({
    "feature": rf_X_test.columns,
    "importance_mean": perm_result.importances_mean,
    "importance_std": perm_result.importances_std
}).sort_values("importance_mean", ascending=False)

display(perm_df.head(15).reset_index(drop=True))

top10_perm = perm_df.head(10).sort_values("importance_mean", ascending=True)

plt.figure(figsize=(10, 7))
plt.barh(top10_perm["feature"], top10_perm["importance_mean"], xerr=top10_perm["
plt.xlabel("Decrease in R² after permutation")
plt.ylabel("Feature")
plt.title("Supplementary permutation importance (top 10)")
plt.show()
```

	feature	importance_mean	importance_std
0	school_stage_Year 6	1.691913	0.150733
1	imd_score	0.321572	0.028476
2	year_start	0.021558	0.003765
3	borough_Southwark	0.005123	0.001429
4	borough_Kensington and Chelsea	0.003587	0.000958
5	borough_Wandsworth	0.002596	0.000689
6	borough_Richmond upon Thames	0.002268	0.000900
7	borough_Hillingdon	0.001392	0.000289
8	borough_Waltham Forest	0.001292	0.000314
9	borough_Westminster	0.001266	0.001059
10	borough_Hounslow	0.000557	0.000308
11	borough_Greenwich	0.000459	0.000184
12	borough_Sutton	0.000458	0.000054
13	borough_Enfield	0.000268	0.000130
14	borough_Hammersmith and Fulham	0.000174	0.000152



Results and discussion

[go back to the top]

5.1 Spatial variation and stage differences

The results of the 2018/19 academic year show that there are obvious spatial differences in childhood obesity prevalence across London boroughs. The average obesity prevalence in Reception is 9.83%, and in Year 6 it is 22.64%, with significant differences between school stages. In all 31 boroughs, prevalence in Year 6 is higher than in Reception, and the average borough-level difference is 12.81 percentage points. This shows that the stage differences are systematic, not only in a few areas. The ranking of each borough also reflects significant inequality. In the 2018/19 academic year, the boroughs with the highest Year 6 obesity prevalence are: Barking and Dagenham, Newham, Enfield, Greenwich and Brent; The lowest are: Richmond upon Thames, Kingston upon Thames, Bromley, Wandsworth, Merton. The map results further confirm this pattern: high-prevalence boroughs are unevenly distributed in London. In general, the above results show that childhood obesity in London has significant spatial structural characteristics.

5.2 Gap patterns over space and time

The analysis of school-stage differences adds an important spatial dimension to the research. Although the obesity rate of Year 6 pupils in all regions is higher than that of Reception children, the magnitude of the difference varies across boroughs. In the 2018/19 academic year, the boroughs with the largest obesity rate gaps are Barking and Dagenham, Newham, Enfield, Westminster, and Kensington and Chelsea; the smallest gaps are in Richmond upon Thames,

Kingston upon Thames, Bromley, Havering, and Sutton. This shows that the inequality between boroughs is not only reflected in the absolute level of obesity rate, but also in the increase in obesity rate from the start to the end of primary school. This gap also shows persistence over time. During the whole study period, the average annual gap was 11.69 percentage points, ranging from 10.01 to 13.20 percentage points. This shows that the obesity rate gap between Reception and Year 6 is a stable feature in the data.

5.3 Deprivation and supplementary modelling

Poverty analysis shows that socio-economic background may explain some of the inequality. The borough-level IMD score is positively correlated with Year 6 obesity prevalence (correlation coefficient ≈ 0.83), and is also positively correlated with the obesity rate gap between Year 6 and Reception (correlation coefficient ≈ 0.72). This means that boroughs with higher deprivation tend to have higher obesity rates and larger school-stage gaps. However, the results need to be interpreted carefully, because the deprivation indicator is treated as a time-invariant background variable, rather than a year-specific measure. The supplementary tree model results generally support the descriptive analysis conclusions. The test-set R^2 of the regression tree is 0.945; the random forest performs slightly better, with a test-set R^2 of 0.972. Feature importance shows that school stage is the strongest predictor, IMD score provides additional explanatory power, and the influence of year is relatively weak.

Conclusion

[go back to the top]

This study shows that childhood obesity in London exhibits significant spatial inequality at the borough level. In the 2018/19 academic year, in all 31 boroughs, the prevalence of obesity in Year 6 was consistently higher than in Reception, and the gap between the two school stages was large and unevenly distributed across space. At the same time, the results show that boroughs with a higher level of deprivation tend to have a higher prevalence of obesity and a larger stage gap, although this study regards deprivation as a static background indicator rather than a year-specific measure. The supplementary tree-based model analysis confirms the important role of school stage and borough context.

References

[go back to the top]

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and regression trees*. Taylor & Francis.

NHS Digital. (2023, October 19). *National Child Measurement Programme, England, 2022/23 school year*. https://www.gov.uk/government/statistics/national-child-measurement-programme-england-202223-school-year

Office for Health Improvement and Disparities. (2023, November 7). *Obesity profile: Statistical commentary November 2023*. https://www.gov.uk/government/statistics/obesity-profile-november-2023-update/obesity-profile-statistical-commentary-november-2023

World Health Organization. (2025, December 8). *Obesity and overweight*. https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight